

Bismuth

Bi (Bi I). Strongest classified transitions below 2 μm (NIST): 18 levels, 39 lines.

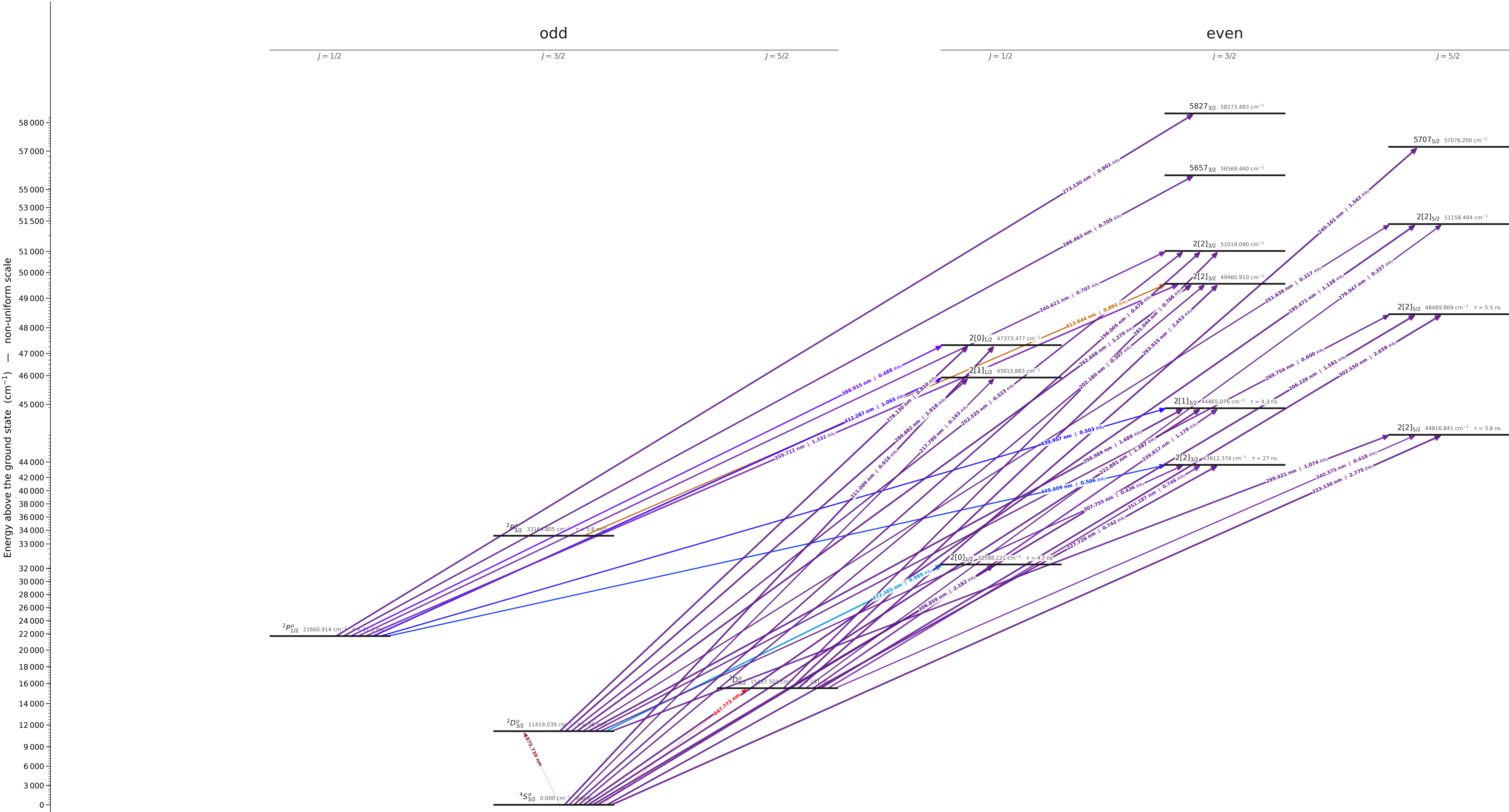
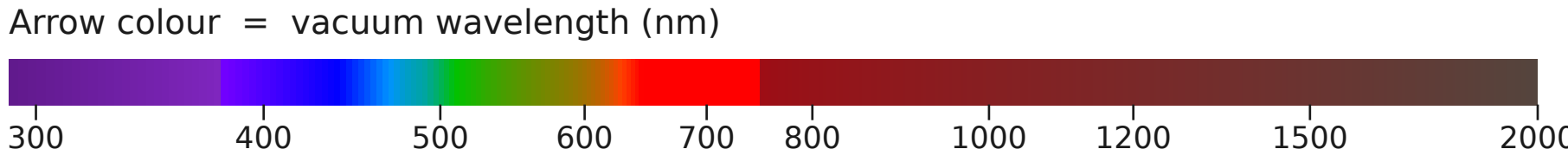
Bi⁺ ionisation limit 58761.65 cm⁻¹ = 7.28552 eV (170.179 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| e r \| J' \rangle|$ in ea_0

Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model or semi-empirical calculation.

Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J \mathbf{er} J' \rangle $ (ea0)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$^4S_{3/2} - 2[0]_{1/2}$	306.8593	306.7702	976.9703	2.182	$1.670e+08$	33.86	78.5	resonance line for collinear and in-source laser spectroscopy
$^4S_{3/2} - 2^2D_{3/2}$	647.7732	647.5943	462.8046	-	-	$1.054e-06$	-	magnetic-dipole line used for parity-nonconservation optical clocks
$^4S_{3/2} - 2^2D_{3/2}$	875.7304	875.4900	342.3342	-	$2.250e+01$	$4.421e-06$	81	magnetic-dipole line of the Oxford parity-nonconservation optical clock

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

$\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

²⁰⁹Bi hyperfine structure ($I = 9/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)					
$4S_{3/2}$	-446.9420(10)	-304.6540(20)	F=3: +3547.64	F=4: +1962.97	F=5: -208.27	F=6: -3093.02		
$2D_{5/2}^o$	-1231.020	-652.500	F=3: +9856.85	F=4: +5367.77	F=5: -651.39	F=6: -8472.51		
$2D_{5/2}^o$	2502.860	23.000	F=2: -34403.78	F=3: -26902.10	F=4: -16897.18	F=5: -4386.71	F=6: +10632.17	F=7: +28162.92
$2P_{3/2}^o$	11260.200	-	F=4: -30965.55	F=5: +25335.45				
$2[0]_{1/2}$	4920.8(6)	-	F=4: -13532.20	F=5: +11071.80				
$2P_{3/2}^o$	491.028	978.640	F=3: -3602.44	F=4: -2290.75	F=5: -39.50	F=6: +3559.10		

Measured A, B constants; sources in the data file.